

Streamlined Design Optimisation of a Novel Organic Solar Cell via Opto-Electrical Modelling

Final Project Report

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Abstract—Organic solar cells (OPVs) are gaining interest as a solar cell technology due to their lightweight and low-temperature solution-based printing advantages over traditional silicon-based solar cells. However, OPV design optimisation often relies on trial and error, which is time-consuming and generates significant material wastage. In response, this paper demonstrates a streamlined computational simulation approach for the design optimisation of a novel OPV architecture, ITO/SnO₂/PTQ10:FCC-Cl/BM-HTL/Ag. The study is split into two stages. First, single cell layer thickness optimisation was performed using an opto-electrical model in OghmaNano. Then, the single cell was upscaled to a module, and module-level design optimisation was performed through a discretised electrical mesh method. The 0.15 cm² single cell achieved a power conversion efficiency (PCE) of 14.60% at optimised layer thicknesses: ITO 100 nm, SnO₂ 50 nm, PTQ10:FCC-Cl 280 nm, BM-HTL 50 nm, and Ag 100 nm. Module design optimisation produced PCE heatmaps to provide practical fabrication guidance for layer thickness, layer offset, stripe width, and number of sub-cells. The optimised module had 100 cm² module area, 5 sub-cells, PTQ10:FCC-Cl thickness 200 nm, and achieved a PCE of 7.25% comparable to current roll-to-roll slot-die coated module performances. The systematic optimisation approach offers practical design insights for scalable module production, reducing development times and material wastage in OPV development.

Index Terms—organic solar cell, organic photovoltaics (OPV), upscaling, design optimisation, opto-electrical modelling.

I. INTRODUCTION

ORGANIC solar cells, also known as organic photovoltaics (OPVs), are a type of solar cell that uses organic semiconductors as the light-harvesting active layer. Their unique advantages, including physical flexibility, lightweight structure, low-cost production, and compatibility with large-scale solution-based printing processes [1], make OPVs a cost-effective alternative to traditional silicon-based solar cells. While challenges such as long-term stability [2] and scalable fabrication [3] remain, recent advances have pushed OPV power conversion efficiency to a record 19.7% [4], positioning them as a promising solution to address the growing global demand for sustainable and renewable energy technologies.

A. Structure of Organic Solar Cells

Organic solar cells can be categorised into single layer, bilayer, bulk heterojunction, ternary, and tandem structures based on the active layer configuration, as shown in Figure 1. Kearns and Calvin [5] reported the first single-layer OPV cell as early as 1958, but its efficiency was extremely low (<0.1%) because the strong Coulomb attraction between electrons and

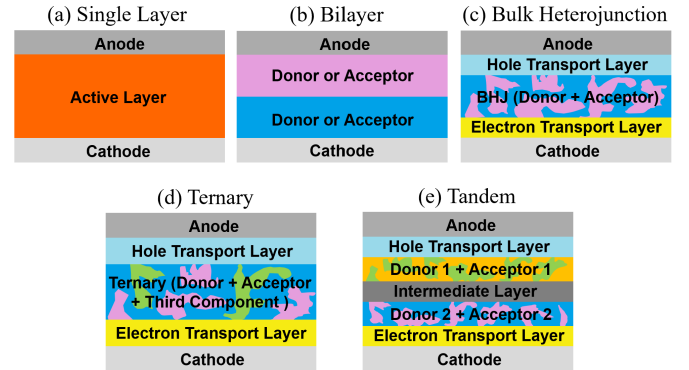


Fig. 1. Schematic diagram of different OPV configurations.

holes in organic materials inhibits electron-hole pair (exciton) dissociation [6]. To overcome this, the active layer must contain at least two organic semiconductors with offset energy levels to enable charge separation at their interface.

As a result, the bilayer OPV was introduced in 1986 by Tang [7]. The bilayer structure improved power conversion efficiency significantly, but it was still only 1% due to inadequate charge transport, because dissociated excitons can only diffuse around 10 nm in organic semiconductors before recombining [6], but a total active layer thickness of above 100 nm is usually required for the OPV to absorb enough light for power generation [6]. Therefore, bilayer cells are either too thin to properly absorb light, or too thick for efficient exciton dissociation.

The solution to this was proposed in 1995 when Yu et al. [8] coined the concept of bulk heterojunction (BHJ), where the donor and acceptor materials are intimately mixed at the nanoscale level to ensure donor-acceptor interfaces are distributed at appropriate diffusion lengths across the entire active layer while maintaining the necessary overall active layer thickness for light absorption. The introduction of the BHJ structure led to a substantial efficiency increase, with the latest world record reaching 19.7% [4], establishing BHJ as a promising photovoltaic technology. Later developments such as ternary [9] and tandem structures [10] have been explored to further enhance performance by adding another donor or acceptor to the active layer mixture or by stacking two active layers, but BHJ has remained the most widely used OPV architecture in the last two decades due to its relative ease of fabrication [9].

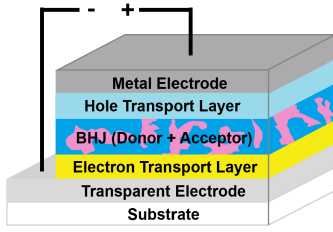


Fig. 2. Schematic diagram of a BHJ OPV device. Light is shone from the bottom, and reaches the active BHJ layer through three transparent layers: the substrate, the transparent electrode, and the ETL.

State of the art BHJ structures are built on a transparent substrate (typically glass or a transparent polymer) coated with a transparent electrode, as illustrated in Figure 2. The transparent electrode is made of a transparent conductive oxide, most commonly indium tin oxide (ITO) due to its excellent transparency and conductivity [11], but recent studies have also looked into alternatives such as metal nano-grids to address ITO's brittle problem [12]. The active BHJ layer is sandwiched between two charge transport layers. The electron transport layer (ETL), typically an oxide like SnO_2 , helps extract and transport electrons from the active layer to the ITO while blocking holes. Conversely, the hole transport layer (HTL), often made of PEDOT:PSS, helps extract and transport holes from the active layer to the metal electrode while blocking electrons. Allowing holes to exit through the metal electrode enables metals that are more resistant to oxygen and moisture (e.g., silver) to be used as the metal electrode, thereby enhancing device stability and performance [13].

OPV performance is commonly measured by open-circuit voltage (V_{OC}), short-circuit current density (J_{SC}), power conversion efficiency (PCE), and fill factor (FF). These parameters are derived from the current density-voltage (J-V) characteristics of a solar cell [9], as illustrated in Figure 3. In dark conditions, the OPV behaves like a diode. Under

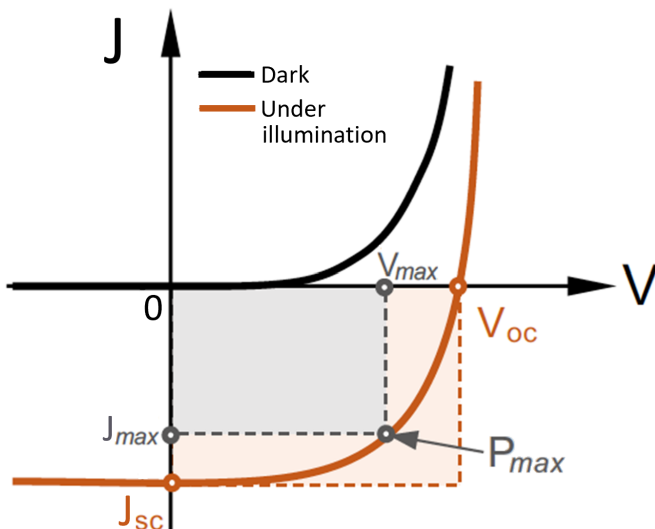


Fig. 3. Typical current density-voltage (J-V) response of a solar cell under dark and illuminated conditions [14].

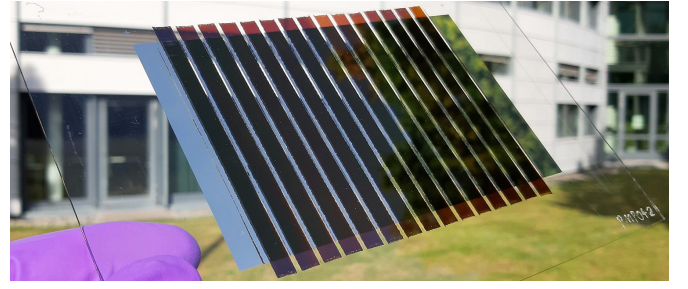


Fig. 4. An OPV module consisting of 15 series-connected cells, where each stripe is an individual cell. Produced by the Fraunhofer Institute for Solar Energy Systems [16].

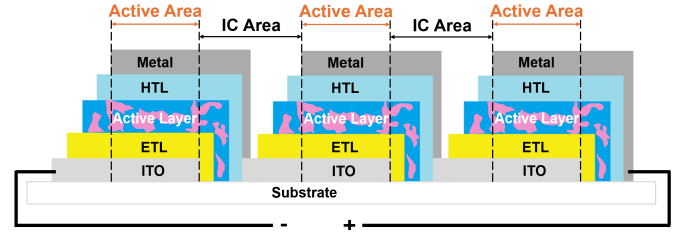


Fig. 5. Cross-section of OPV module stack, showing the active and interconnection (IC) areas, and positive and negative terminals.

illumination, power is produced, where PCE is defined as the ratio of maximum produced power (P_{max}) to the incident light power. Fill factor quantifies how closely the device's output characteristics approach the ideal:

$$FF = \frac{P_{max}}{V_{OC} \times J_{SC}} = \frac{V_{max} \times J_{max}}{V_{OC} \times J_{SC}} \quad (1)$$

However, V_{OC} of a single OPV cell is typically below 1 V due to the fundamental energy difference between the highest occupied molecular orbital (HOMO) of the donor and the lowest unoccupied molecular orbital (LUMO) of the acceptor [15]. To meet the voltage requirements of practical electronic applications, OPVs are typically fabricated as modules with multiple cells connected in series, allowing their individual voltages to sum linearly, as depicted in Figures 4 and 5. This structure is commonly called a module stack.

In the module stack, each layer is intentionally offset to improve manufacturing reliability. First, it simplifies series connection by naturally connecting the metal electrode of one cell to the ITO electrode of the next. Second, it reduces the risk of short circuits by minimising the likelihood of pinholes or defects overlapping across layers [17]. Finally, it helps mitigate edge-related defects caused by misalignment in directly stacked layers [17].

From a design perspective, the module stack also introduces a functional distinction between the active area, where photovoltaic conversion takes place, and the interconnection (IC) area, which links cells in series but does not generate power. The geometric fill factor (GFF) quantifies the proportion of these areas [18]:

$$GFF = \frac{\text{Active Area}}{\text{Active Area} + \text{IC Area}} \quad (2)$$

A high GFF indicates that a larger proportion of the module actively generates power, while a low GFF means more inactive 'dead space', leading to low overall PCE even if individual cells are highly efficient. Therefore, maximising GFF is often a key to achieving high module performance, which requires not only careful design but also precise deposition, as the choice of fabrication method directly affects layer morphology, layer alignment, and scalability [19].

B. Fabrication and Upscaling

Module fabrication uses scalable printing techniques that can be classified based on their patterning abilities [19]. Zero-dimensional (0D) techniques, such as blade coating and spray coating, have no patterning ability and are only used for creating smooth films across the entire substrate surface. One-dimensional (1D) methods like slot-die coating can deposit stripes, while two-dimensional (2D) techniques, including screen printing and inkjet printing, can deposit complex 2D patterns. Additionally, versatile techniques like gravure and flexographic printing can deposit both 1D and 2D patterns. These methods are often designed for roll-to-roll production that can reach speeds of several metres per minute [20].

TABLE I
HIGHEST REPORTED PCE BY PRINTING METHODS

Printing Method	PCE (%)	Device Area (cm ²)	Module	Group [Ref.]
Blade coating	14.46	204	38 cells	Basu et al. [21]
Slot-die coating	9.57	30	5 cells	Lee et al. [22]
Spray coating	2.44	4	3 cells	Huang et al. [23]
Screen printing	2.38	3	3 cells	Zhang et al. [24]
Gravure printing	1.02	45	5 cells	Yang et al. [25]

At the moment, blade coating, slot-die coating, spray coating, screen printing, and gravure printing have been explored in the literature for OPV module fabrication. Table I summarises the highest reported module PCEs achieved using each method. It is clear that blade coating leads with a record PCE of 14.46%. However, as a 0D technique, it cannot produce patterned module stacks without additional etching steps [26], which limits its throughput and scalability.

As a result, slot-die coating stands out as the more practical choice for large-scale OPV fabrication. This method operates by continuously dispensing a liquid solution through a narrow slit (the "slot-die") onto a moving substrate, making it well-suited for fabricating striped module stacks and enabling high-speed printing ($>10 \text{ m min}^{-1}$) [20].

Although there is still an efficiency gap between slot-die printed OPVs and silicon-based solar cells, advances in materials, device architectures, and fabrication techniques have steadily improved OPV performance, and their low energy demand during production means their energy payback period may be as short as one day [27]. As a result, this has driven growing research interest in scalable OPV modules [3] aimed at pushing OPVs one step closer to commercial viability.

However, a key obstacle in this transition is that OPV design optimisation still relies on trial-and-error, which is time-consuming, generates significant material wastage, and

often struggles to identify the optimal OPV configuration due to the complex interplay of multiple parameters. This highlights the need for systematic, simulation-based strategies to streamline the optimisation process.

Some studies have already demonstrated the potential of computational simulation in addressing parts of the OPV design. For example, Basu et al. [21] employed computational fluid dynamics (CFD) to optimise the coating speed for a 204 cm² module, as well as a finite element method (FEM) model to determine the optimal number of sub-cells for the 204 cm² module with a fixed IC width of 135 μm .

Other approaches have used genetic algorithms to optimise interfacial layer thickness [28], but the computational cost of this method limits it to just one or two design layers. Meanwhile, Cao et al. [29] combined Design of Experiments (DoE) with machine learning to optimise donor-acceptor ratio and solution concentration, but their work focused solely on small-area single cell optimisation. These computational approaches either target specific fabrication parameters or address only isolated aspects of the design layout. A streamlined simulation strategy for design optimisation from a single cell up to a module remains largely unexplored.

C. Aims and Objectives

In response, this project demonstrates a streamlined design optimisation approach using the OPV simulation software OghmaNano [30] [31] [32] to optimise a novel BHJ OPV structure glass/ITO/SnO₂/PTQ10:FCC-Cl/BM-HTL/Ag, as shown in Table II. This configuration is developed by Swansea-based OPV demonstrator SPECIFIC, and achieves 12.77% PCE in laboratory testing of a 0.15 cm² single cell.

TABLE II
NOVEL OPV DEVICE STRUCTURE

Material	Thickness (nm)	Resistivity ($\Omega \text{ m}$)
ITO	100	1.50×10^{-6}
SnO ₂	50	5.71×10^{-6} [33]
PTQ10:FCC-Cl	150-200	1347.89
BM-HTL	50	0.025 [34]
Ag	100	1.59×10^{-8}

The design optimisation follows a two-step process. First, an opto-electrical model is built in OghmaNano to assess how layer thickness affects single cell performance, aiming to identify optimal thicknesses based on simulations and manufacturing constraints. Next, the design is upscaled to a module using a 3D electrical mesh to evaluate how layout parameters influence overall performance in order to maximise PCE within SPECIFIC's roll-to-roll slot-die coating limits. The project provides simulation-based design recommendations to the SPECIFIC fabrication team to support scalable OPV module development and saves both time and material costs by reducing reliance on trial-and-error testing.

The models and simulation setup are detailed in Section II. Results are presented in Section III, split into single cell layer thickness optimisation in Section III-A and module layout optimisation in Section III-B. Slot-die fabrication constraints

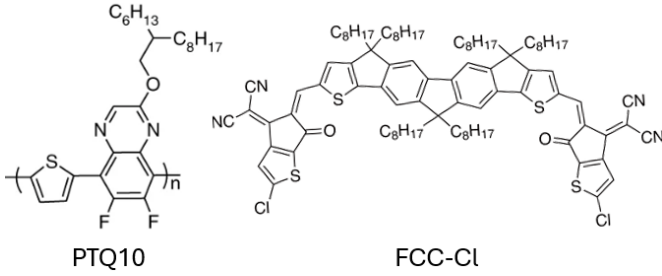


Fig. 6. Chemical structures of PTQ10 [35] and FCC-Cl [36].

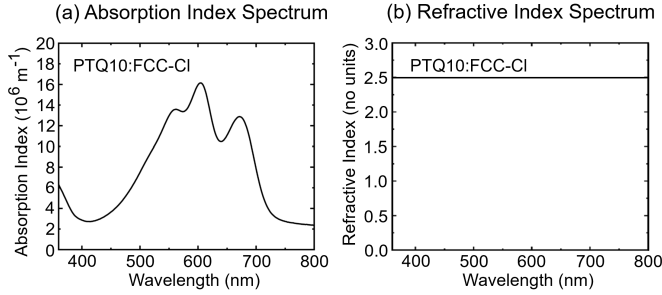


Fig. 7. Absorption and refractive index spectra of PTQ10:FCC-Cl.

are also discussed with informed feedback from the SPECIFIC team. Lastly, the conclusion is provided in Section IV.

II. METHODOLOGY

This project uses the OPV simulation tool OghmaNano developed by Dr. Roderick MacKenzie [30] [31] [32]. It employs a 1D/2D drift-diffusion model [31] and explicitly solves the Shockley-Read-Hall equations [31] to account for carrier trapping which are significant in disordered materials such as organic semiconductors. OghmaNano stores simulation data in JSON format, making it straightforward to systematically vary OPV layout parameters through Python scripting.

Each OPV model in OghmaNano is described by an optical model, which describes light absorption and distribution in the OPV structure, and an electrical model, which models the OPV as a series of diodes and resistors. The optical model is unanimous as long as the materials in an OPV configuration stays the same. However, the electrical model is available in two variants: a 1D model for single-cell simulations and a 3D model for module-level simulations. The optical model is explained in Section II-A, the 1D electrical model in Section II-B, and the 3D electrical model in Section II-C.

A. Optical Model

Light absorption, reflection, and transition within an OPV are modelled in OghmaNano by the transfer matrix method [31], which calculates the distribution of light within the OPV structure as a function of position and wavelength, taking into account reflections from the interfaces and optical loss within the structure. To accurately calculate light distribution, this method requires the absorption and refractive index spectra of

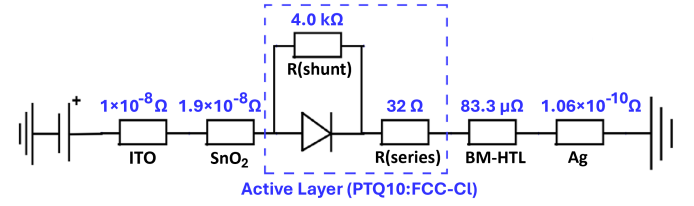


Fig. 8. Circuit diagram for the 1D electrical model.

TABLE III
FINE-TUNED DIODE PARAMETERS

Parameter	Value
Photon efficiency	0.759
Ideality factor	1.727
I_0	1×10^{-13} A

each material as input. For the glass/ITO/SnO₂/PTQ10:FCC-Cl/BM-HTL/Ag device under investigation, glass, ITO, SnO₂, and Ag are directly available from OghmaNano's built-in material database. However, the active layer PTQ10:FCC-Cl and the hole transport layer BM-HTL are custom materials that had to be manually added.

The active layer, PTQ10:FCC-Cl, is a blend of the donor material PTQ10 [35] and the acceptor material FCC-Cl [36]. Their chemical structures are shown in Figure 6. This is a novel blend and has not been previously studied as an active layer for OPVs. Its absorption index spectrum was provided by SPECIFIC, while its refractive index spectrum was assumed constant at 2.5, as no laboratory or literature data were available. This value was selected based on an analysis of the refractive index spectra of organic semiconductors in OghmaNano's database and simulation testing, where a value of 2.5 was found to yield reliable simulation results. The final absorption and refractive index spectra for PTQ10:FCC-Cl, as used throughout this project, are shown in Figure 7.

BM-HTL, a material manufactured by Brilliant Matters [34], does not have published absorption and refractive index spectra. As a result, the spectra for PEDOT:PSS, a common HTL available in OghmaNano's database, were used as a substitute. This substitution is unlikely to significantly affect the simulation, as BM-HTL is positioned after the active layer so its optical absorption is not expected to impact charge generation, and the HTL layer is typically very thin (around 50 nm). The layer type is set as "Active" for the active layer and "Other" for the rest of the layers.

B. 1D Electrical Model

A single cell OPV is modelled using the 1D electrical model based on the circuit diagram in Figure 8, where the active layer is modelled as a diode with a series resistance $R(\text{series})$ and a shunt resistance $R(\text{shunt})$ to account for resistive and leakage losses, whereas the non-charge generating layers are modelled as resistors. The glass substrate is omitted in the electrical model because it is not part of the electrical circuit.

Resistors and diode parameters are modelled to match experimental data provided by SPECIFIC for a 0.15 cm² OPV cell at the layer thicknesses recorded in Table II. Resistances

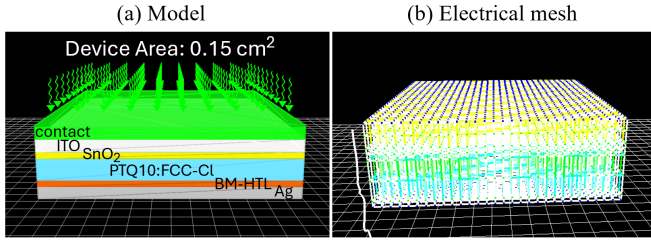


Fig. 9. 3D model and corresponding electrical mesh to replicate the 0.15 cm² single cell.

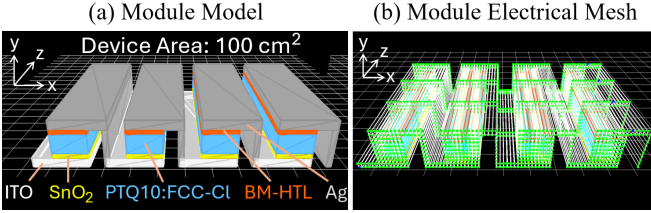


Fig. 10. 3D model of a 4-cell OPV module and corresponding electrical mesh. Electrical contacts are made at the left and right ends of the module.

for the ITO, SnO₂, BM-HTL, and Ag layers are calculated using their known bulk resistivity values [33] [34] by applying Ohm's law. Then the active layer parameters including diode parameters, $R(\text{series})$, and $R(\text{shunt})$ are fine-tuned so that the simulated J-V curve matches experimental J-V measurements. Figure 8 and Table III show the optimised active layer parameters, which reproduce the experimental V_{OC} , J_{SC} , and PCE values with less than 0.5% error, see Figure 11.

Once fitted to experimental data, the model can simulate single cell performance at varying layer thicknesses, which are detailed in Section III-A.

C. 3D Electrical Model

Simulating a large-area OPV module in OghmaNano requires generating a 3D electrical mesh based on the 1D circuit model to discretise the OPV module so that the governing equations of optical interaction, charge transport, and recombination can be solved numerically across the entire structure.

As an initial step, the 0.15 cm² single cell is replicated using the 3D mesh to verify that its performance metrics remain consistent with the 1D model, as shown in Figure 9. The 3D mesh is made up of nodes and segments, where each segment represents a diode with series and shunt losses for the active layer, or a resistor for the ITO, SnO₂, BM-HTL, and Ag layers. The same parameter values from the 1D model are used, with the only difference being that photon efficiency is reduced to 0.661 for the 3D model to account for additional surface reflection as part of the OghmaNano 3D mesh setup. The 3D model matches experimental data with less than 0.9% error, as shown in Figure 11.

Next, the 0.15 cm² 3D mesh is upscaled into a 100 cm² module akin to the example shown in Figure 10. The module model is tailored to include only the active area and relevant electrical connections, omitting minute structural details in the IC area as these do not contribute to power generation. Module

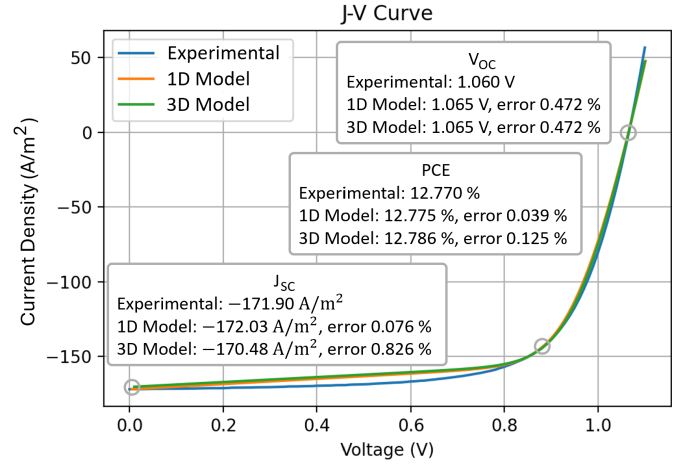


Fig. 11. Best fit between experimental and simulated J-V curves.

length in the z-direction does not affect V_{OC} , J_{SC} , FF, and PCE because they are measured in per unit area. Therefore, the only relevant parameters that influence module performance are in the x- and y-directions, which are detailed in Section III-B.

III. RESULTS AND DISCUSSION

A. Single-Cell

Layer thickness is the primary parameter that determines light absorption, charge generation, and series resistance in single cells. In contrast, parameters like cell area mostly influence ease of fabrication, not the intrinsic efficiency of a single cell.

The 0.15 cm² single cell was simulated using the 1D model under AM1.5 illumination at 1000 W/m². The thickness of ITO, SnO₂, PTQ10:FCC-Cl, BM-HTL, and Ag were varied from 10 nm to 800 nm in 5 nm increments to determine OPV performance trend with changing layer thickness. With each variation of layer thickness, layer resistances were automatically adjusted according to Ohm's law through Python scripting. Power conversion efficiencies were calculated from simulated J-V curves and presented in Figure 12.

• ITO - Figure 12(a)

ITO shows a decreasing trend in PCE with increasing thickness, likely due to optical interference and transmission losses. Since light must pass through the transparent ITO to reach the active layer, thicker ITO layers may reflect or absorb more light, reducing the amount of light reaching the active layer. In addition, ITO has a relatively high resistivity ($1.50 \times 10^{-6} \Omega m$), so thicker ITO layers can exacerbate resistive losses and reduce PCE. Therefore, a thin ITO layer is generally preferred for transparency and low resistive losses. However, practical thickness is limited by fabrication constraints, because very thin ITO films (< 50 nm) are difficult to deposit and can face uniformity and connectivity issues [37].

ITO is usually deposited on glass substrates via magnetron sputtering, as this technique ensures clean, precise deposition with excellent optoelectronic properties [37]. As a result, ITO is typically not deposited in-house, but is instead sourced from

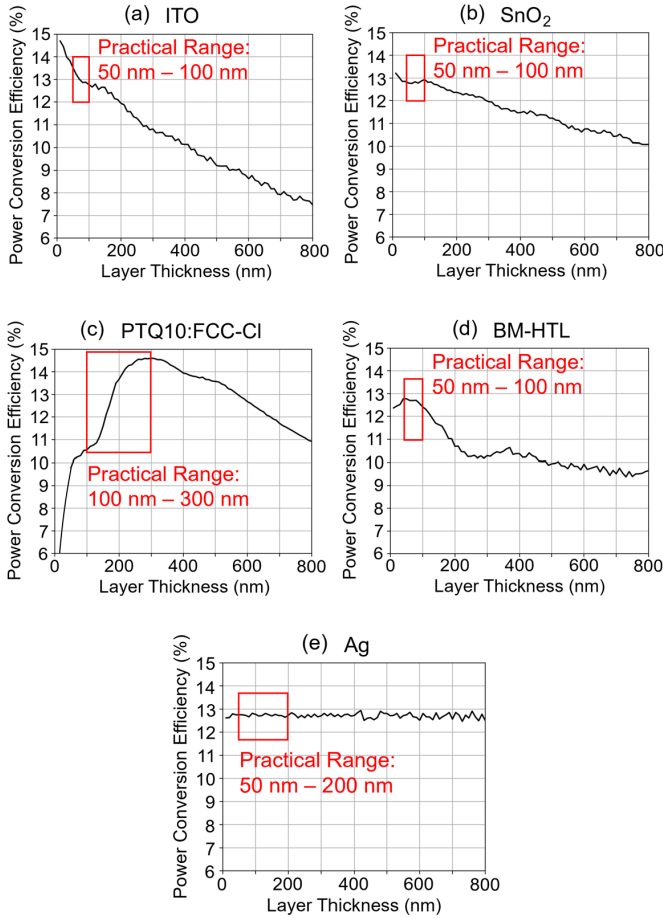


Fig. 12. Simulated single cell PCE trends for varying thicknesses of each functional OPV layer: ITO, SnO₂, active layer (PTQ10:FCC-Cl), BM-HTL, and Ag. Red boxes highlight practical thickness ranges based on fabrication constraints and material considerations.

commercial suppliers as glass substrates pre-coated with ITO thin films. These are most commonly supplied with a 100 nm ITO layer, though some suppliers like Techinstro offer 50 nm alternatives [38]. Therefore, a practical ITO thickness range of 50–100 nm is generally recommended for device fabrication.

• SnO₂ - Figure 12(b)

SnO₂ also shows a decreasing PCE trend with increasing thickness due to optical and resistive losses, so a thin SnO₂ layer is generally preferable. However, this project is concerned with upscaling the single cell into a module using slot-die coating, so minimum SnO₂ thickness is constrained by the minimum printable thickness (around 50 nm) [39], but layer thicknesses up to 100 nm are sometimes adopted to ensure film quality and uniformity. Therefore, practical SnO₂ thickness range is set at 50–100 nm.

• PTQ10:FCC-Cl - Figure 12(c)

The active layer thickness directly controls how much light is absorbed. Very thin active layers (e.g., 50 nm) do not absorb enough photons so photogenerated current is low, whereas thicker layers absorb more photons and generate more excitons. As a result, PCE increases with thickness up to an optimal point, beyond which it declines due to saturating light absorption and inefficient charge extraction

caused by recombination and resistive losses. Figure 12(c) found the optimal PCE at 300 nm, which is notably higher than the typical 80–120 nm [40] reported for most OPVs. This unusually high optimum may be attributed to two factors. First, prior studies [36] [41] have shown that both PTQ10 and FCC-Cl exhibit high tolerance to film thickness, maintaining performance even at thicknesses up to 300 nm. Second, the absorption index spectrum of PTQ10:FCC-Cl beyond 800 nm wavelength remains unknown, and the refractive index spectrum was approximated due to a lack of literature data, potentially introducing uncertainties in the optical simulations. Considering these factors, the practical thickness range for PTQ10:FCC-Cl is set at 100–300 nm.

• BM-HTL - Figure 12(d)

BM-HTL again shows a declining PCE trend with rising film thickness likely due to series resistive losses, while light absorption is an unlikely concern because BM-HTL is positioned after the active layer. Therefore, a thin BM-HTL layer is generally preferred, but slot-die coating limits the minimum achievable thickness to around 50 nm [39], and thicker layers up to 100 nm are sometimes used to ensure film quality and uniformity. As a result, the practical thickness range is set at 50–100 nm.

• Ag - Figure 12(e)

Lastly, the thickness of the silver layer has minimal impact on PCE due to its excellent conductivity. Therefore, the primary goal is to achieve a thick, uniform silver contact, but this is typically kept below 200 nm due to the high cost of silver. The lower limit is again set at 50 nm due to slot-die coating limitations. Consequently, practical thickness range is set at 50–200 nm, with SPECIFIC opting for 100 nm.

• Grid Search and Top Performance

A grid search was performed for the practical thickness ranges, and the maximum PCE was found to be 15.54% at ITO 50 nm, SnO₂ 50 nm, PTQ10:FCC-Cl 280 nm, BM-HTL 50 nm, and Ag 100 nm, with $V_{OC} = 1.07$ V, $J_{SC} = 221.23$ A/m², and fill factor = 0.65. When ITO was increased to 100 nm, maximum PCE drops to 14.60% and J_{SC} to 209.59 A/m², while V_{OC} and fill factor remain the same to two decimal places. Although this is below the current world record of 19.7% [4], it is still competitive with PCEs commonly achieved in OPVs in the last five years (13% to 15%) [21], suggesting that it is a promising OPV design and a viable candidate for integration into scalable module manufacturing.

B. Module

Design optimisation of OPV modules focuses on four parameters: (1) layer thickness, (2) layer offset, (3) stripe width, and (4) number of cells in the module, as labelled in Figure 13. In general, $l_1 = l_2 = l_3 = l_4$ and $w_1 = w_2 = w_3 = w_4 = w_5$ are used for ease of fabrication unless otherwise stated. These parameters were selected for their direct influence on light absorption, charge transport, GFF, and resistive losses that are critical for determining module-level PCE. They were systematically analysed to evaluate module performance, and the

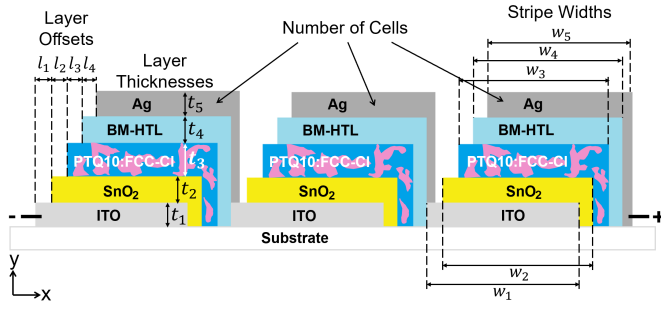


Fig. 13. Cross-section of OPV module stack highlighting key dimensional features.

results are presented in dedicated subsections, providing both simulated performance trends and practical considerations for scalable module fabrication.

• Layer Thickness - Figure 14

Although layer thickness was previously investigated for a single cell in Section III-A, transitioning to a module introduces factors such as lateral resistive losses that may alter its performance trends. Therefore, re-evaluating layer thickness in the context of a full module is necessary.

In response, a multi-cell module based on SPECIFIC's existing design ($l_1 = 1.5$ mm, 87×100 mm module area) was simulated, varying layer thickness from 50 to 400 nm (up to 500 nm for PTQ10:FCC-Cl) in 5 nm steps. Simulations were run for 2 to 6 sub-cells to assess if PCE trends hold across different configurations. Figure 14 shows the resulting PCEs, labelling practical thickness ranges based on performance and slot-die coating constraints.

Looking at individual layers, ITO shows increasing PCE with thickness in modules with fewer cells, but the trend plateaus or decreases as cell count rises. This behaviour differs from that observed in the single cell, likely due to a change in the dominant current flow. In a single cell, vertical current flow dominates in the ITO, so thicker films have higher resistance and lower PCE. In contrast, in a module, horizontal (lateral) current flow dominates in the ITO, so thicker films have lower resistance and higher PCE. As a result, in modules with fewer (i.e. wider) cells, current travels longer lateral current paths in the ITO, so lateral resistance has a marked impact, and PCE increases with ITO thickness. In modules with more (i.e. narrower) cells, shorter lateral current paths means the effect of lateral resistive losses is less pronounced and vertical resistance and optical losses become more influential, causing PCE to decline with ITO thickness.

To reflect the changed ITO trend, practical ITO thickness range is adjusted to 100-200 nm. The rest of this paper uses ITO = 100 nm to match SPECIFIC's standard practice and ensure the results are relevant to their design needs. Nonetheless, simulations indicate that increasing ITO thickness to 200 nm may improve PCE by up to 2.23% depending on the number of cells in the module, suggesting a potential opportunity for performance gain.

SnO₂ and BM-HTL thicknesses appear to have minimal impact on overall PCE, as vertical current flow dominates in these layers and their vertical resistive losses are small

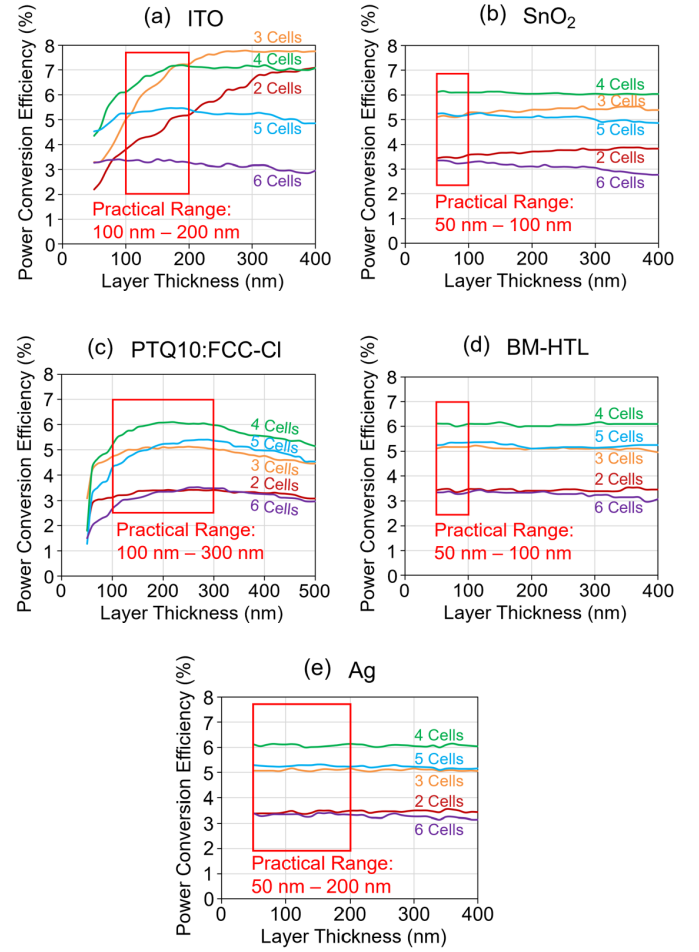


Fig. 14. Simulated module PCE trends for varying thicknesses of each functional OPV layer: ITO, SnO₂, active layer (PTQ10:FCC-Cl), BM-HTL, and Ag. Red boxes highlight practical thickness ranges based on fabrication constraints and material considerations.

compared to lateral losses in the ITO. Ag thickness also shows little effect on PCE likely due to its low resistivity ($1.59 \times 10^{-8} \Omega \text{ m}$). Therefore, practical thickness ranges in these layers follow those from the single cell. The rest of this paper adopts 50 nm for SnO₂, 50 nm for BM-HTL, and 100 nm for Ag considering slot-die coating constraints and material use.

The active layer (PTQ10:FCC-Cl) has a similar trend to that observed in the single-cell, but PCE appears to plateau earlier (between 180 nm and 280 nm), indicating reduced thickness sensitivity at module scale. Across all module configurations tested, peak PCE was consistently observed near 200 nm, which supports its recommendation as a reliable active layer thickness for diverse module configurations. Therefore, the rest of this paper uses 200 nm for PTQ10:FCC-Cl thickness.

Among the module configurations tested for the 87 cm² module, the 4-cell design offers the best performance likely due to a balance between ITO lateral resistive losses and GFF. Modules with fewer cells (e.g., 2) tend to experience higher ITO lateral resistive losses, leading to reduced fill factor and lower overall PCE. On the other hand, modules with many cells (e.g., 6) have lower lateral resistive losses, but frequent IC regions can reduce GFF to as low as 0.18, which means

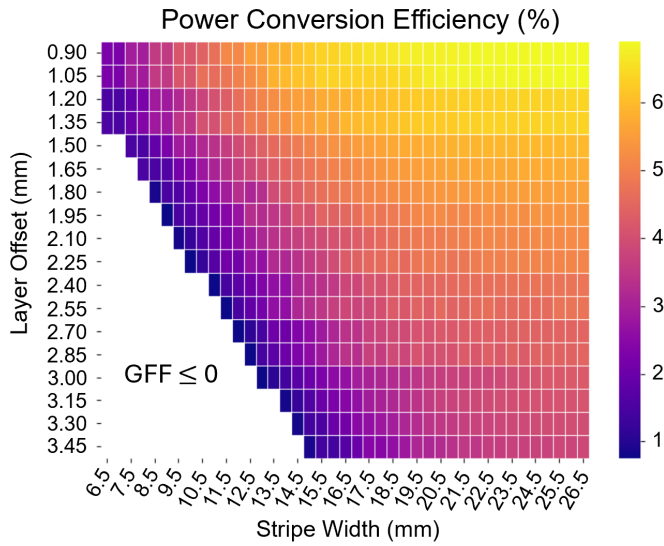


Fig. 15. Power conversion efficiency heatmap for a 4-cell module under varying layer offset and stripe width at PTQ10:FCC-Cl thickness 200 nm. White regions indicate invalid configurations which result in negative or zero GFF.

only 18% of the module area is actively generating power. At an intermediate cell count (e.g., 4), these trade-offs are balanced: resistive losses are moderate, and the GFF remains relatively high, yielding the highest overall PCE.

Compared to a single cell, however, the best-performing 4-cell module's PCE (6.08% at ITO = 100 nm, SnO₂ = 50 nm, PTQ10:FCC-Cl = 200 nm, BM-HTL = 50 nm, and Ag = 100 nm) is still less than half that of the single cell. This is likely caused by the combined effect of GFF and ITO lateral resistive losses. The 4-cell module has a GFF of 0.48, which means only 48% of its area is actively generating power, so module PCE is expected to drop to 7.01% due to GFF alone, and lateral resistance likely contributes to the remaining PCE losses.

As a result, optimising GFF and reducing lateral resistive losses are essential for minimising PCE losses during upscaling. Both can be improved by minimising layer offset, which reduces IC area and the ITO width outside the active area.

• Layer Offset and Stripe Width - Figure 15

Layer offset and stripe width were jointly analysed for a 4-cell module to evaluate module performance. The resulting PCEs are presented as a heatmap in Figure 15, showing that smaller layer offsets and wider stripes generally yield higher PCE, likely because smaller layer offsets reduce IC area and wider stripes increase active area. This effect is evident in the heatmap's upper-right region, where layer offsets below 1 mm and stripe widths above 21 mm achieve PCEs over 6.8% and GFFs above 0.72, showing a 0.72% increase from the previous layer thickness analysis.

However, while wider stripes improve simulated PCE, in practice they are more prone to printing defects like pinholes [17] that cannot be quantitatively evaluated without laboratory testing. Therefore, layer offset is prioritised for further optimisation due to its clearer performance impact and well-defined fabrication limits based on printing resolution.

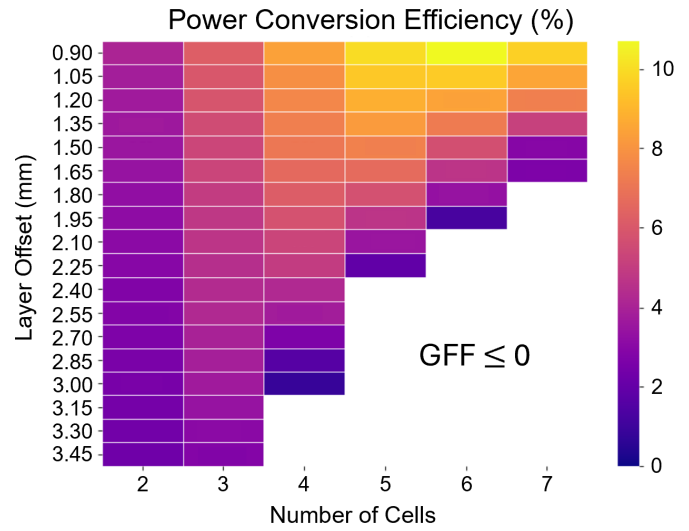


Fig. 16. Power conversion efficiency heatmap for a 100 cm² module under varying layer offset and number of cells at PTQ10:FCC-Cl thickness 200 nm. White regions indicate invalid configurations which result in negative or zero GFF.

• Layer Offset and Number of Cells - Figure 16

Having identified layer offset as a key parameter for performance optimisation, the next stage of analysis explores how it interacts with another important design choice: the number of cells within a fixed-width module. In practice, the overall module size is often limited by application-specific constraints such as substrate width, meaning that efficiency improvements must come from internal design optimisation. Therefore, a second heatmap analysis was conducted to evaluate different combinations of layer offset and number of cells on a fixed-width module, as illustrated in Figure 16. A module area of 100 × 100 mm was used in this analysis, as it represents the maximum substrate size compatible with SPECIFIC's roll-to-roll manufacturing process. All simulations used layer thicknesses at ITO = 100 nm, SnO₂ = 50 nm, PTQ10:FCC-Cl = 200 nm, BM-HTL = 50 nm, and Ag = 100 nm.

Consistent with earlier results, smaller layer offsets increase PCE, even in a fixed-width module. For example, in a 6-cell module, reducing the layer offset from 1.05 mm to 0.90 mm can increase PCE by up to 1.15%.

Regarding the number of cells, PCE peaks at 5–6 cells before declining, showing the same trade-off between lateral resistive losses and GFF previously discussed in the layer thickness analysis (Figure 14). The best configurations achieve PCEs ranging from 8.46% to 10.71%, representing a significant improvement over the 6.08% observed in the earlier analysis.

• SPECIFIC Feedback and Response - Figure 17

At this stage, feedback was received from SPECIFIC regarding the practical feasibility of the simulated layer offset values. Although simulations suggest that peak PCE is achieved at the lowest possible layer offset (down to 0.90 mm in the heatmaps), such small values are not practically achievable in SPECIFIC's manufacturing process. While their slot-die coating equipment offers a width resolution of approximately

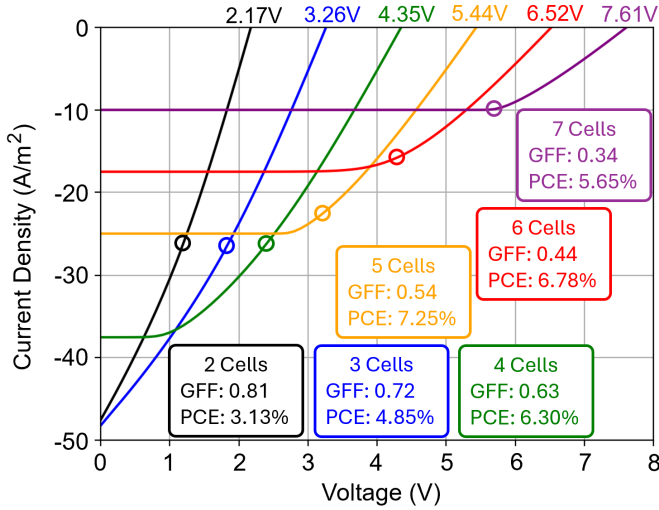


Fig. 17. JV curves for different number of cells on a 100 cm² module at $l_1 = 2$ mm and $l_2 = l_3 = l_4 = 1$ mm, illustrating GFF and PCE trends. Circles indicate the locations of maximum power.

100 μ m, lateral movement of the substrate during roll-to-roll printing introduces further alignment variability. Consequently, the realistically achievable minimum layer offsets are estimated to be $l_1 = 2$ mm and $l_2 = l_3 = l_4 = 1$ mm. Based on these values, J–V curves were generated for varying number of cells for the 100 cm² module, labelled with their corresponding GFF and PCE, as shown in Figure 17.

The J–V curves show that V_{OC} increases approximately linearly with cell count as expected from a series-connected module. J_{SC} decreases with increasing number of cells because GFF declines and reduces the effective light-absorbing area of the module. Fill factor improves steadily from 0.31 at 2 cells to 0.74 at 7 cells, likely due to reduced lateral resistive losses in the ITO as stripe widths narrow. PCE reaches its maximum at a balance point between lateral resistive losses and GFF, which occurs at 5 cells and achieves GFF = 0.54, $V_{OC} = 5.44$ V, $J_{SC} = 25.01$ A/m², fill factor = 0.53, and PCE = 7.25%. The optimised module has GFF losses of 46% and resistive losses of 4.34%.

In comparison, Basu et al. [21] reported a module with PCE of 14.46% at a GFF of 0.96, with GFF losses of 3.49% and resistive losses of 1.91%. However, this exceptionally high GFF was only achieved because they used blade coating followed by an etching step to pattern the narrow 135 μ m IC widths [21]. This method, despite its high precision, has low throughput and is not compatible with roll-to-roll manufacturing, where high-throughput techniques like slot-die coating are required.

The highest reported PCE for a slot-die coated OPV module is 9.57% with total losses of 29.1% [22]. However, GFF was not reported in that study, making it difficult to assess how much of the loss stemmed from GFF or lateral resistance. While the 7.25% PCE achieved in this work falls short of that record, it still lies at the upper end of the typical 5–7% PCE range for roll-to-roll slot-die printed modules [22]. Moreover, this result was based on SPECIFIC’s realistic manufacturing constraints, using practically achievable layer offsets and with-

out relying on post-processing steps such as etching. These findings demonstrate the strength of the simulation framework and the potential of the optimised design for scalable OPV production.

While this study is entirely simulation-based, it is important to acknowledge several practical factors that may cause deviations between simulated and fabricated module performance. Firstly, fabrication processes inherently introduce variability such as thickness variation during layer deposition, so experimental PCE may fluctuate by up to $\pm 1\%$ even under controlled conditions. Secondly, in series-connected modules, the total current is limited by the lowest-performing cell, so any mismatch in cell performance due to non-uniform layer thicknesses, material inconsistencies, or local defects can reduce overall PCE. Therefore, the specific fabrication conditions must be considered when interpreting the simulation results and translating them into practical design guidelines.

Looking forward, this work serves as a foundation for experimental validation and further refinement of module designs under real-world constraints. Future efforts could focus on collecting more accurate optical data for the absorptance and refractive index spectra of PTQ10:FCC-Cl and BM-HTL, as well as obtaining more experimental J–V curves to improve the reliability of the simulated average device performance. Finally, building and testing real modules based on the design recommendations would help validate simulation results and reveal any fabrication challenges that may not have been captured in simulations, which could help make informed improvements to the simulation framework.

IV. CONCLUSION

This paper demonstrated a simulation-based approach to streamline the design optimisation of a 0.15 cm² OPV device with a configuration of glass/ITO/SnO₂/PTQ10:FCC-Cl/BM-HTL/Ag. The study was split into two steps. First, an opto-electrical model was built in OghmaNano to optimise layer thickness in the 0.15 cm² single cell. The optimised single cell achieved a PCE of 14.60% at layer thicknesses of ITO = 100 nm, SnO₂ = 50 nm, PTQ10:FCC-Cl = 280 nm, BM-HTL = 50 nm, and Ag = 100 nm, comparable with typical OPV performance.

Next, the single cell was upscaled into a multi-cell module using a 3D electrical mesh. Module layout optimisation was performed for four key structural parameters: layer thickness, layer offset, stripe width, and number of cells. Based on SPECIFIC’s practical fabrication limitations, the optimum layer thicknesses were found at ITO = 100 nm, SnO₂ = 50 nm, PTQ10:FCC-Cl = 200 nm, BM-HTL = 50 nm, and Ag = 100 nm, and the best-performing design was a 5-cell, 100 cm² module with a GFF of 0.54 and a PCE of 7.25%, demonstrating the performance trade-off between GFF and lateral resistive losses, and putting it at the higher end of typical OPV module PCEs. These findings provide actionable design guidelines for scalable OPV module fabrication at SPECIFIC, thereby reducing development times and material wastage.

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